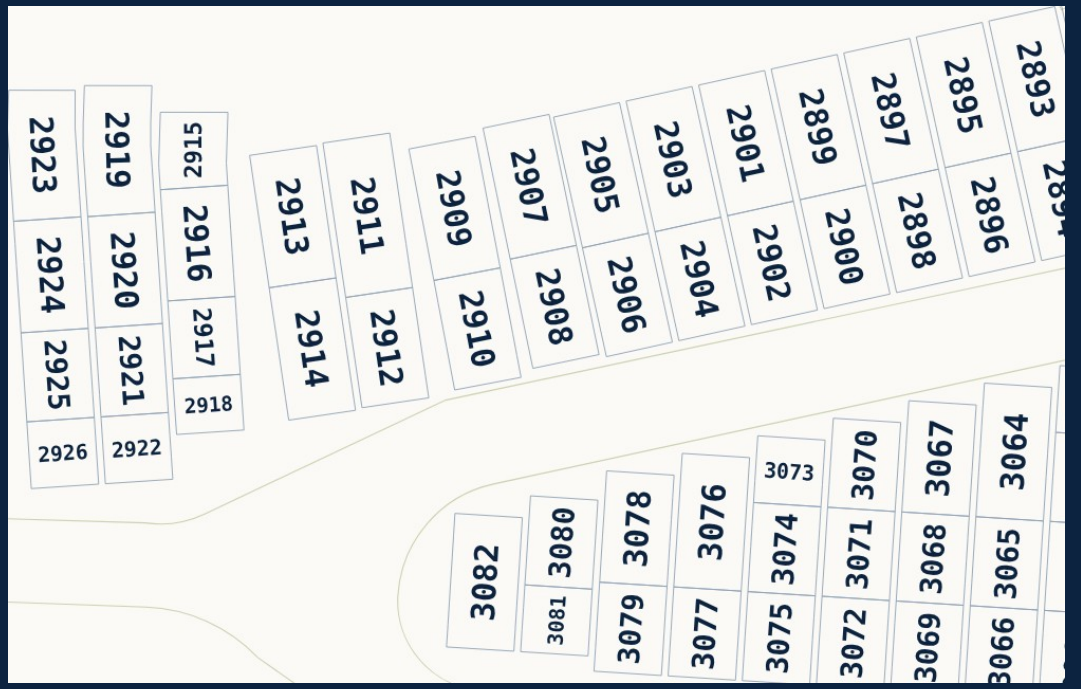
Cedar Grove H-Number Atlas

Progress report: the map, the numbers behind it, and what comes next

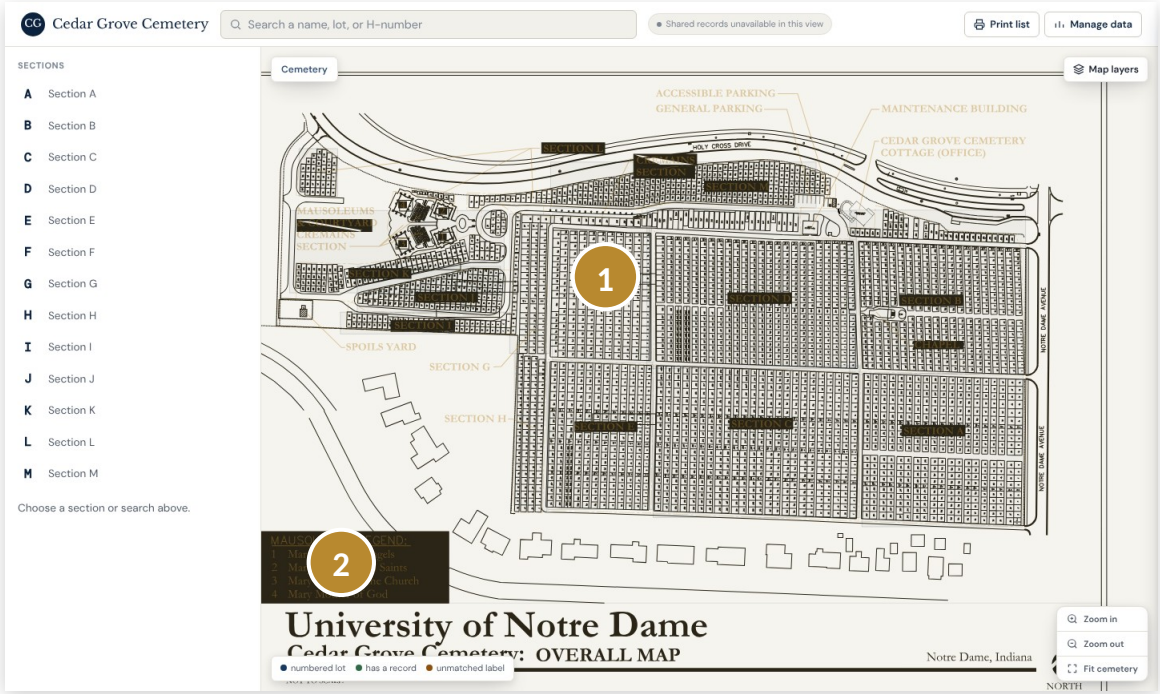


Current map, curved lot rows (H002891 to H003082)

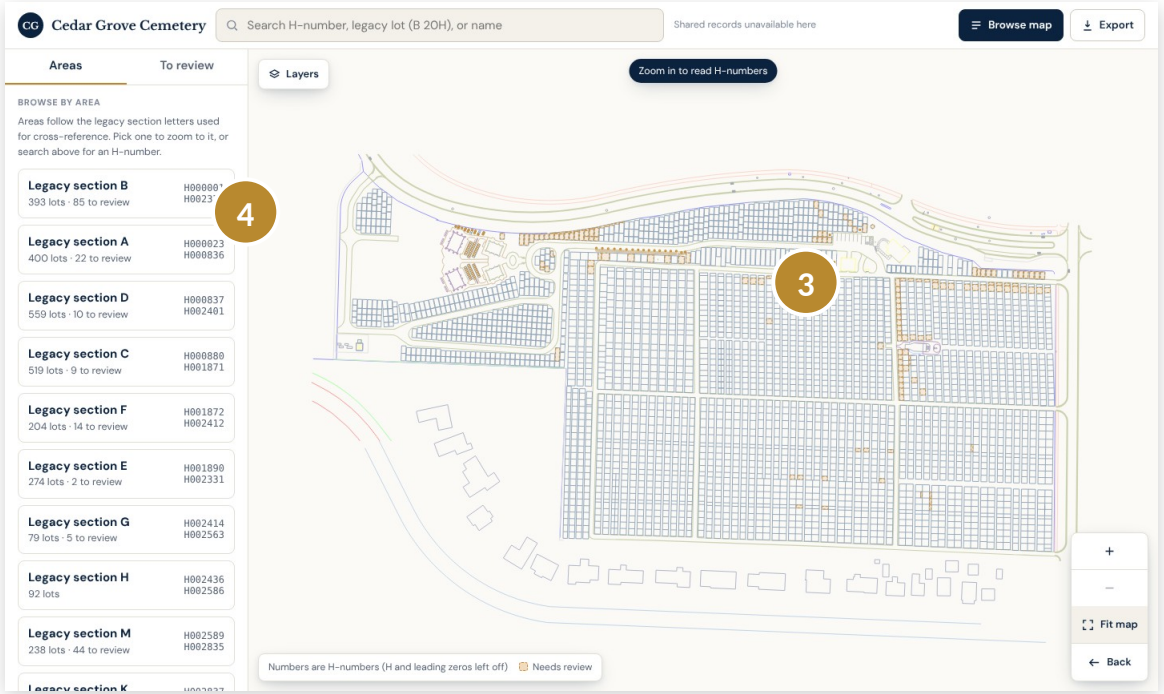
University of Notre Dame, Cedar Grove Cemetery | October 2026

The whole map is cleaner and easier to read

BEFORE: previous site



NOW: current site



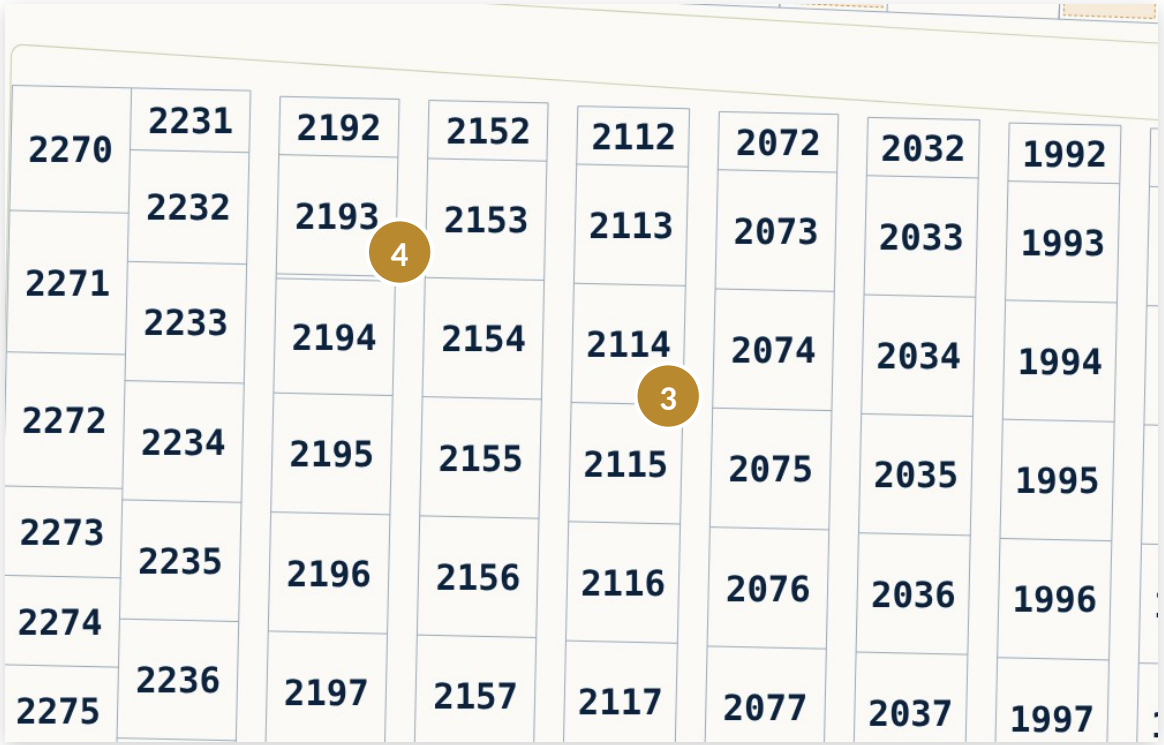
1. Dark "SECTION" title blocks covered lots. They are gone.
2. Legend, parking and office notes are gone from the default map.
3. Roads, buildings and every lot outline are kept, in lighter lines.
4. Search, areas with H-number ranges, and a list of lots to review.

Each lot shows one number, centered inside it

BEFORE Same area and zoom, east side of the main grid



NOW



1. Old numbers still show underneath the new ones.
2. Heavy outlines crowd the numbers against the lot ends.
3. One H-number per lot, centered, with clear space around it.
4. Thin outlines; no old numbers anywhere.

Curved and angled lots keep their numbers inside

BEFORE Same area and zoom, curved rows



NOW



1. A "SECTION" block sits over lots 2894 to 2909.
2. Numbers run across lot edges; black smudges are erased old labels.
4. Each number turns with its lot and stays fully inside it.
5. Narrow lots get a smaller number, never a clipped one.

Where things stand: what is mapped, unmatched, and still missing

On the map

3,103

lots verified: drawn outline, H-number and legacy label all agree

Unmatched

218

visible lots whose number is unresolved

68

decoder lots with no map position

Information still missing

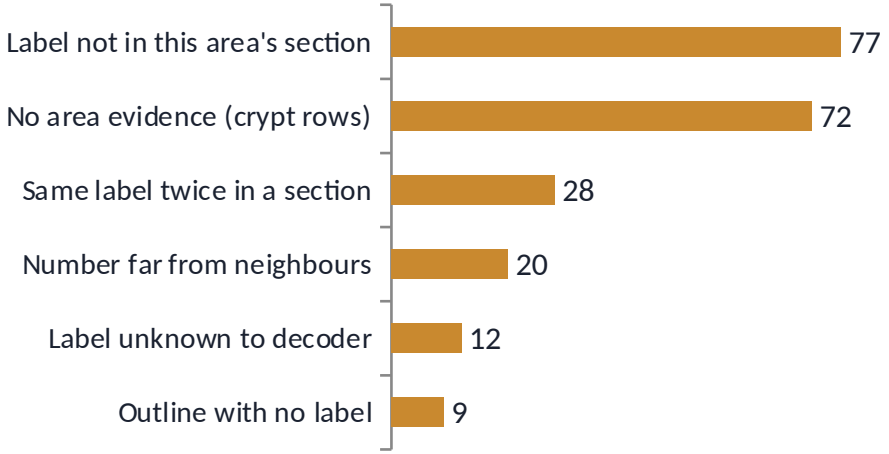
0

burial records entered so far

How the 3,103 were confirmed



Why the 218 are unresolved



0

lots with a verified grave count

Unresolved lots are shaded, never numbered by guesswork. A candidate H-number appears only in the lot panel.

Previous build, for reference (project figures): 3,063 lots placed, 136 unplaced entries listed, 49 unmatched labels. The categories are not identical, so the counts are shown side by side, not as a percentage.

How the map is built, and why each step exists

1 Original drawings

Two PDFs: an overall map and nine section pages. Lot numbers are drawn as tiny shapes, not text.

Now: the CAD file holds real text, so numbers are read directly.

2 Glyph recognition

Software matched each drawn digit against templates for 0 to 9, A, B and H.

Now: kept only as an independent cross-check.

3 Coordinate registration

Section pages were lined up on the overall map (error under 0.05 map units).

Now: old readings are aligned to the CAD file (0.1 unit).

4 H-number decoding

The Python decoder turns "Section B, lot 20H" into H000001. It is the only source of H-numbers.

Now: unchanged.

5 SVG rendering

The map is drawn as scalable shapes, so it stays sharp at any zoom and prints cleanly.

Now: each number is measured and fitted inside its lot.

6 Site and PDFs

The interactive site and the printable PDFs are generated from the same data.

Now: rebuilt, with lot panels and review states.

Tools and technical work, in plain terms

Python 3

Runs the whole build, from the drawing to the finished site, with one command.

PyMuPDF

Reads the lines and text out of the drawing, and writes the PDFs.

OpenCV, NumPy, SciPy

Find each lot's outline and match positions between the two drawings.

Shapely

Geometry maths that keeps every number inside its own lot.

s4decoder1.py

The decoder: legacy lot label in, H-number out.

SVG and plain JavaScript

The interactive map. Stays sharp when zoomed and needs no add-ons.

Playwright and Chromium

Take the screenshots and run the automatic checks in a real browser.

Shared database (Artifact)

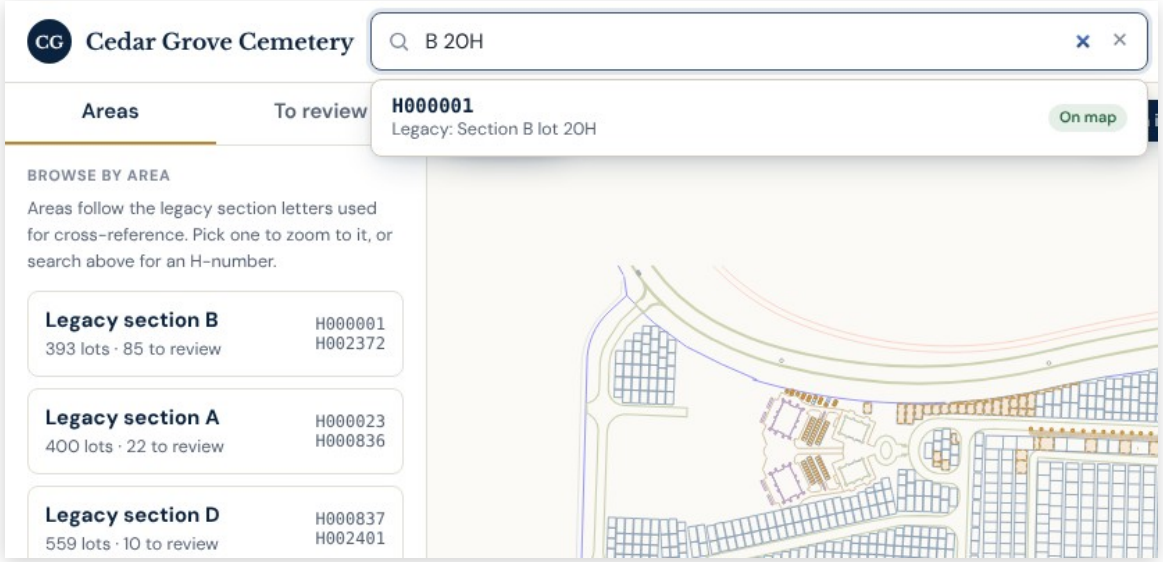
Stores grave records. Only people with edit access can write.

Data formats: .ai / PDF (source drawing) > JSON (working data) > SVG (map), PDF (print), CSV (lists and imports). Excel (.xlsx) import is built but not yet tested.

Automatic checks: every number lies inside its lot (3,103 of 3,103); 0 of 773,021 number pixels touch a lot line on screen, and 0 of 1.74 million in the PDF.

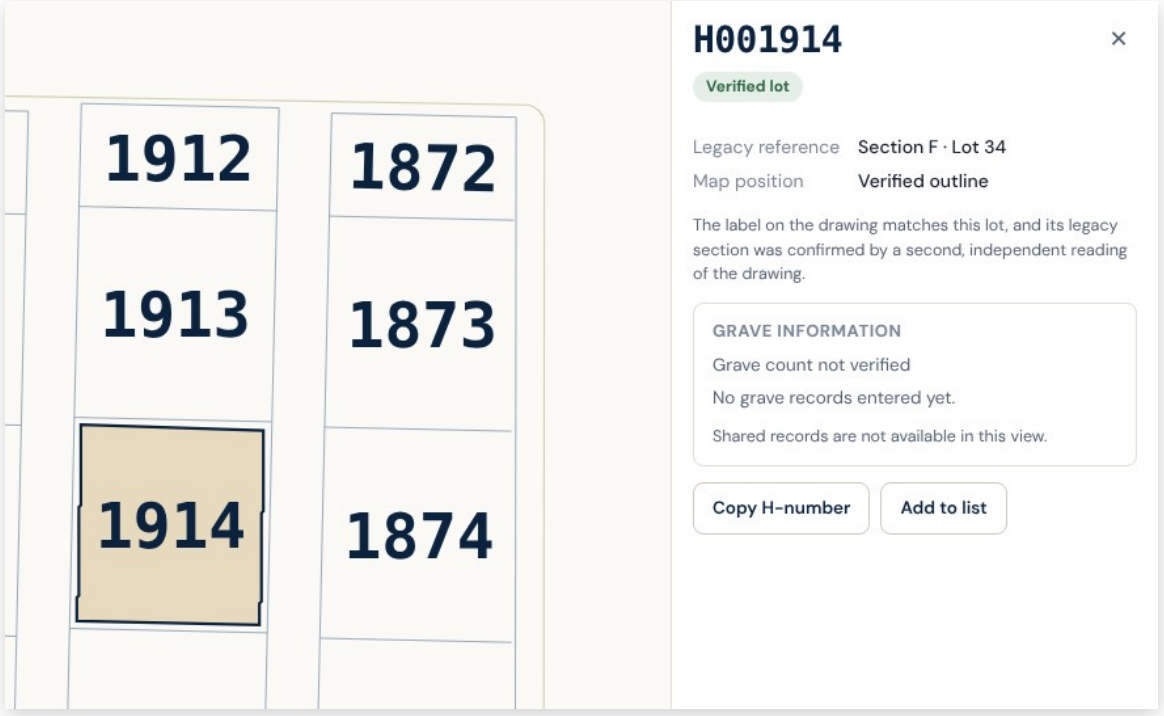
Website: find a lot and open it

Working: tested in a browser



Search by H-number, by legacy lot such as "B 20H", or by recorded name (names appear once records exist). Close matches are suggested while typing.

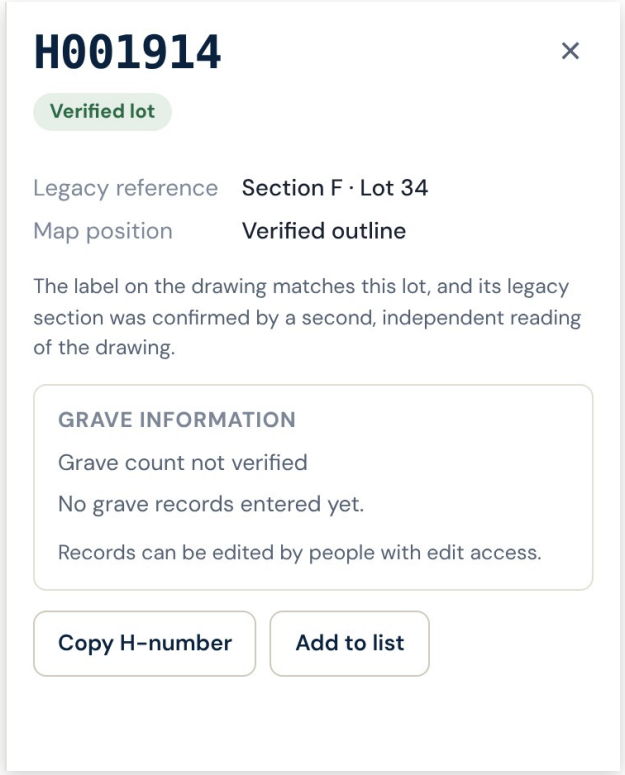
Working: tested in a browser



Click any lot and the map zooms to it, highlights it and opens its panel beside the map, so the lot stays visible. Back and Fit map return to earlier views.

Website: records, editing and exports

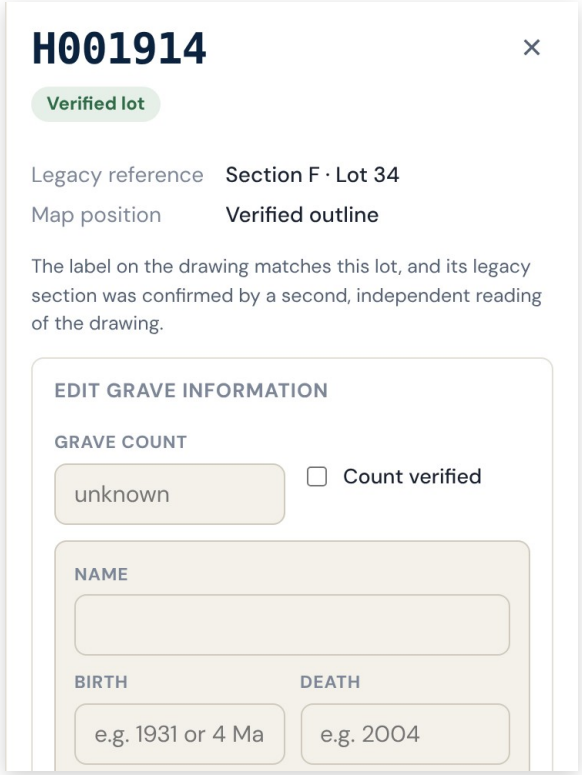
Viewers



Working

Fields start empty and say so. No guessed graves. No Edit button.

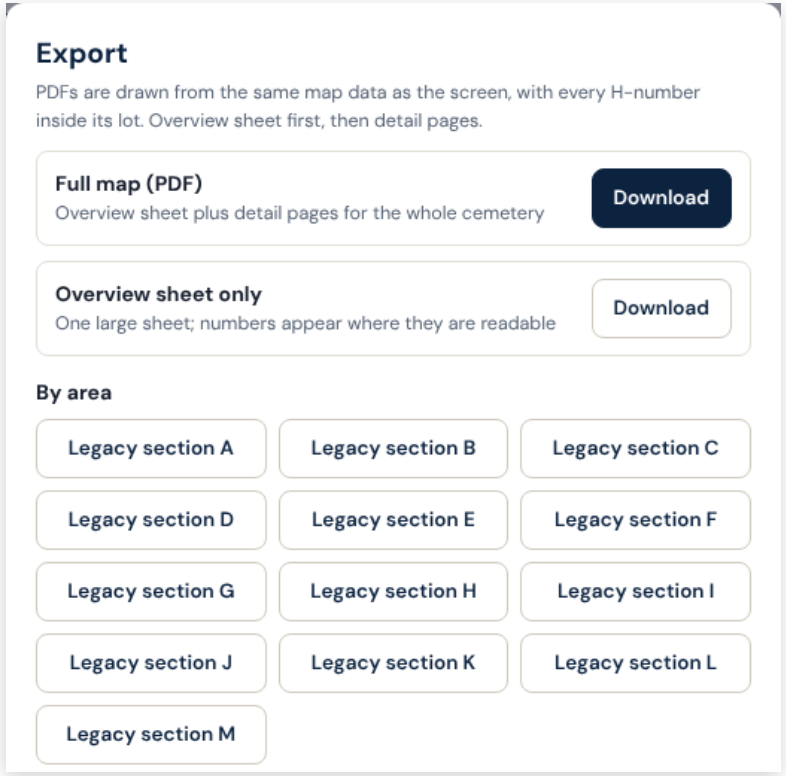
Editors



Not yet tested live

Edit button and form only for people with edit access. Tried with a simulated sign-in. Import is built, untested.

Exports



PDFs, CSV built **Download untested** **Excel: planned**

Full map and each area as PDF, plus CSV lists. Excel (.xlsx) export is planned; CSV opens in Excel.

What happens next, in priority order

What needs to happen	What we need from cemetery staff
1. Settle the numbering conflicts Section B 373 to 380H appear in two places. Section G has lot 19 twice. Section A lots 20H to 343H show as plain numbers on the map. Several B-side labels (395 to 414) are unknown to the decoder.	Someone who knows the original numbering to say which version is right for each conflict.
2. Finish the map corrections 72 crypt-row labels have no section. 9 outlined lots have no label. 3 lots are too small for a number (E 195A, E 195B, H 21).	Which numbering the crypt rows follow, and a site check of the 9 unlabelled lots.
3. Verify grave counts per lot Lots hold different numbers of graves, so the map shows "grave count not verified" until it is confirmed.	A per-lot capacity sheet: lot, number of graves, and how they are arranged.
4. Provide the master record spreadsheet One row per burial, matched to a lot by H-number or by section and lot. The import is ready.	Columns: H-number or section + lot, name, birth, death, grave position, source, notes.
5. Test live and decide Try editing, import and downloads with real accounts. Settle area names, logo and where the site is hosted.	Editor accounts for testing, and a decision on area names and hosting.